

//Address Example 1

```
// Include the iostream library for input and output operations
#include <iostream>
// Use the standard namespace for convenience
using namespace std;
// Define the main function
int main(){
// Declare an integer variable x and initialize it with the value 25
    int x = 25;
// Print the address of memory for variable x using the address-of operator (&)
    cout << "The address of memory for x is " << &x << endl;
// Return 0 to indicate successful execution of the program
    return 0;
}
```

```
The address of memory for x is 0x7fffe737a86c
```

//Address Example 2

```
#include <iostream>
using namespace std;
int main(){
    int x = 25;
    int y = x;
    cout << "y=" << y << endl;
    cout << "The address of memory for x is " << &x << endl;
    cout << "The address of memory for y is " << &y << endl;
}
```

```
y=25
The address of memory for x is 0x7fff6ce4ed8c
The address of memory for y is 0x7fff6ce4ed88
```

//Reference Example 1

```
#include <iostream>
using namespace std;
int main(){
    int x = 25;
    int &y = x;
    cout << "The value of x is " << x << endl;
    cout << "The value of y is " << y << endl;
    cout << "The address of memory for x is " << &x << endl;
    cout << "The address of memory for y is " << &y << endl;
}
```

```
The value of x is 25
The value of y is 25
The address of memory for x is 0x7ffcc4e68c5c
The address of memory for y is 0x7ffcc4e68c5c
```

//Address Example 3

```
#include <iostream>
using namespace std;
int main(){
    int x = 25;
    int &y = x;
    cout << "The value of x is " << x << endl;
    cout << "The value of y is " << y << endl;
    cout << "The address of memory for x is " << &x << endl;
    cout << "The address of memory for y is " << &y << endl;
    y = 50;
    cout << "The value of x is " << x << endl;
    cout << "The value of y is " << y << endl;
    cout << "The address of memory for x is " << &x << endl;
    cout << "The address of memory for y is " << &y << endl;
}
```

```
The value of x is 25
The value of y is 25
The address of memory for x is 0x7ffd36dd530c
The address of memory for y is 0x7ffd36dd530c
The value of x is 50
The value of y is 50
The address of memory for x is 0x7ffd36dd530c
The address of memory for y is 0x7ffd36dd530c
```

//Reference Example 2

```
#include <iostream>
using namespace std;
int main(){
    int x = 25;
    int &y = x;
    cout << "The value of x is " << x << endl;
    cout << "The value of y is " << y << endl;
    cout << "The address of memory for x is " << &x << endl;
    cout << "The address of memory for y is " << &y << endl;
    y = 50;
    cout << "The value of x is " << x << endl;
    cout << "The value of y is " << y << endl;
    cout << "The address of memory for x is " << &x << endl;
    cout << "The address of memory for y is " << &y << endl;
    int z = 100;
```

```

y = z;
cout << "The value of x is " << x << endl;
cout << "The value of y is " << y << endl;
cout << "The value of y is " << y << endl;
cout << "The address of memory for x is " << &x << endl;
cout << "The address of memory for y is " << &y << endl;
cout << "The address of memory for y is " << &z << endl;
}

```

```

The value of x is 25
The value of y is 25
The address of memory for x is 0x7ffde0179f3c
The address of memory for y is 0x7ffde0179f3c
The value of x is 50
The value of y is 50
The address of memory for x is 0x7ffde0179f3c
The address of memory for y is 0x7ffde0179f3c
The value of x is 100
The value of y is 100
The value of y is 100
The address of memory for x is 0x7ffde0179f3c
The address of memory for y is 0x7ffde0179f3c
The address of memory for y is 0x7ffde0179f38

```

//Address Error Example

```

#include <iostream>
using namespace std;
int main(){
    int x = 25;
    int y = x;
    int z = &x;
    cout << "z=" << z << endl;
    cout << "The address of memory for x is " << &x << endl;
    cout << "The address of memory for y is " << &y << endl;
    cout << "The address of memory for z is " << &z << endl;
}

```

```

./main.cpp:6:9: error: cannot initialize a variable of
type 'int' with an rvalue of type 'int *'
   6 |     int z = &x;
     |           ^
1 error generated.
make: *** [Makefile:10: main] Error 1

```

//Address Example 4

```

#include <iostream>
using namespace std;
int main() {
    int x = 25;
    int y = x;
    int* z = &x;
}

```

```
cout << "The value of x is " << x << endl;
cout << "The address of memory for x is " << &x << endl;
cout << "The value of y is " << y << endl;
cout << "The address of memory for y is " << &y << endl;
cout << "The value of z is " << z << endl;
cout << "The address of memory for z is " << &z << endl;
cout << "The value of *z=" << *z << endl;
return 0;
}
```

```
The value of x is 25
The address of memory for x is 0x7ffc7dd04c6c
The value of y is 25
The address of memory for y is 0x7ffc7dd04c68
The value of z is 0x7ffc7dd04c6c
The address of memory for z is 0x7ffc7dd04c60
The value of *z=25
```